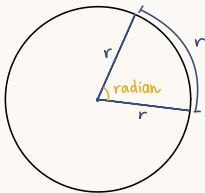


PURE 4. Circular measure



• The **arc** of the angle **1 radian**
= length of **radius**

• circumference = $2\pi r$
↳ $360^\circ = 2\pi$ radians

Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

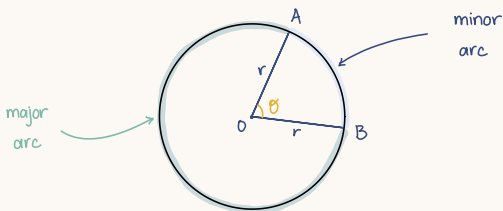
* conversion:

• $a^\circ \rightarrow \text{rad} : a \times \frac{\pi}{180^\circ}$

• $b \text{ rad} \rightarrow ^\circ : b \times \frac{180^\circ}{\pi}$

(* $1 \text{ rad} = 1 \times \frac{180^\circ}{\pi} \approx 57^\circ$)

A) Arc length

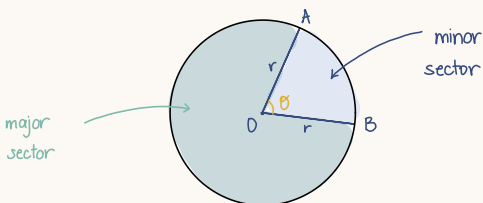


• when θ measured in **radians**

↳ **length** of AB = $r\theta$

(arc length = $\frac{\theta}{2\pi} \times 2\pi r = r\theta$)

B) Area of a sector



• when θ measured in **radians**

↳ **area** of sector AOB = $\frac{1}{2} r\theta^2$

(area = $\frac{\theta}{2\pi} \times \pi r^2 = \frac{1}{2} r^2 \theta$)